LEARNQUEST

LINUX LABS DOCUMENTATION

MODULE 4

Searching, Viewing, Editing & Processing Text

Command-Line Practice Documentation

Linux Fundamentals Coursework

July 2026

Introduction

The Linux command line is the foundation for navigating, managing, and troubleshooting any Linux-based system. This module brings together practical, hands-on commands used to search through files, view and edit their contents, gather quick statistics, and transform text — the everyday building blocks every Linux user, administrator, or developer relies on. Rather than presenting commands as a single unordered list, this documentation groups them by purpose, pairs each command with a clear explanation, and includes terminal screenshots so the syntax and output can be seen exactly as they appear in practice.

Learning Objectives

- Search and filter text inside files using pattern-matching with grep, including anchors, negation, and extended regular expressions.
- View, create, overwrite, and append file content using cat and ls, and understand the difference between the >, >>, and << redirection operators.
- Generate quick file statistics — line, word, and character counts — using wc, and combine it with other commands through piping.
- Navigate and edit files directly in the terminal using the nano and vi text editors.
- Apply sed to perform stream-based find-and-replace operations and redirect the results into new files.
- Understand how pipes (|) and redirection operators chain simple commands together into more powerful workflows.

Learning Path

1. Start with Searching & Filtering — learn to locate patterns inside files with grep before manipulating them.
2. Move to Viewing & Writing Files — practice reading, overwriting, and appending content with cat and ls.
3. Learn File Statistics — use wc to measure file size and combine it with pipes for filtered output.
4. Explore Text Editors — get comfortable editing files interactively using nano and vi.
5. Finish with Text Processing — use sed to automate find-and-replace edits across files.

Command Reference by Category

Each category below lists the relevant commands with a short explanation, followed by the terminal screenshot(s) demonstrating that command in action.

1. Searching & Filtering — grep

Locating patterns and lines of text inside files.

Command	Description
grep daemon.*nologin /etc/passwd	Searches /etc/passwd for the line that ends with “nologin”, matching anything before it starting with “daemon”.
grep ^root /etc/passwd	Searches /etc/passwd for the line that starts with “root”.

Command	Description
grep -v nologin$ /etc/passwd	Displays only the lines that do NOT end with “nologin” (inverted match with -v).
grep -E "root|coder" /etc/passwd	Uses extended regular expressions (-E) to find lines containing either “root” or “coder”.

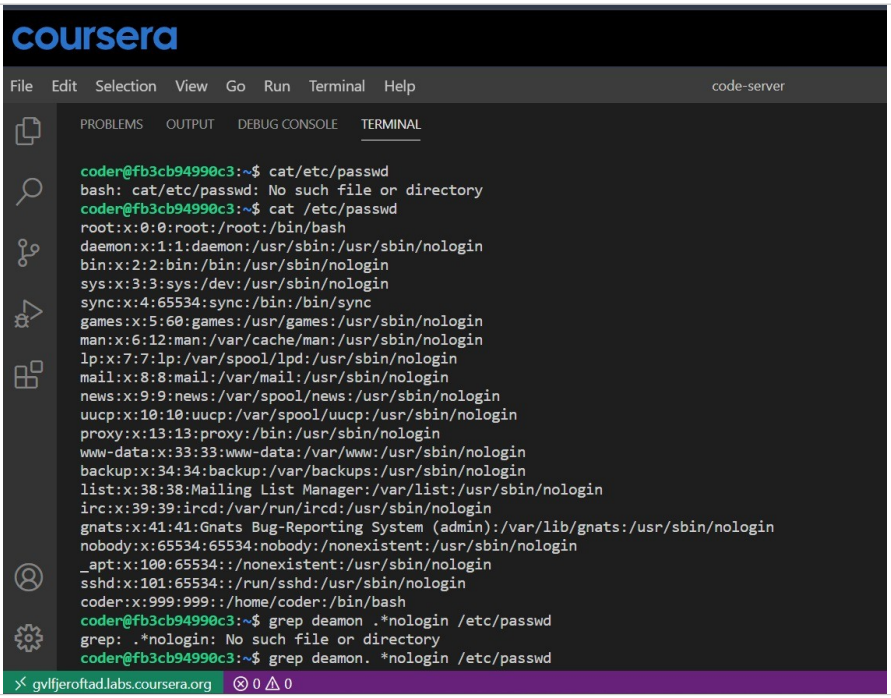

Figure 1.1 — grep searching /etc/passwd by pattern and anchor.

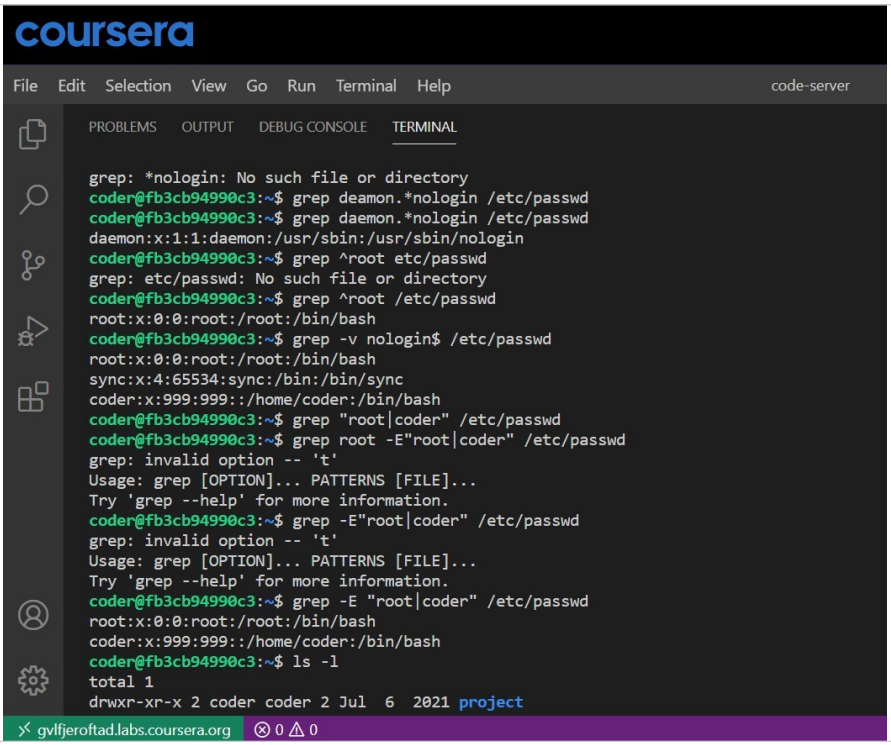

Figure 1.2 — grep -v excluding lines that match a pattern.

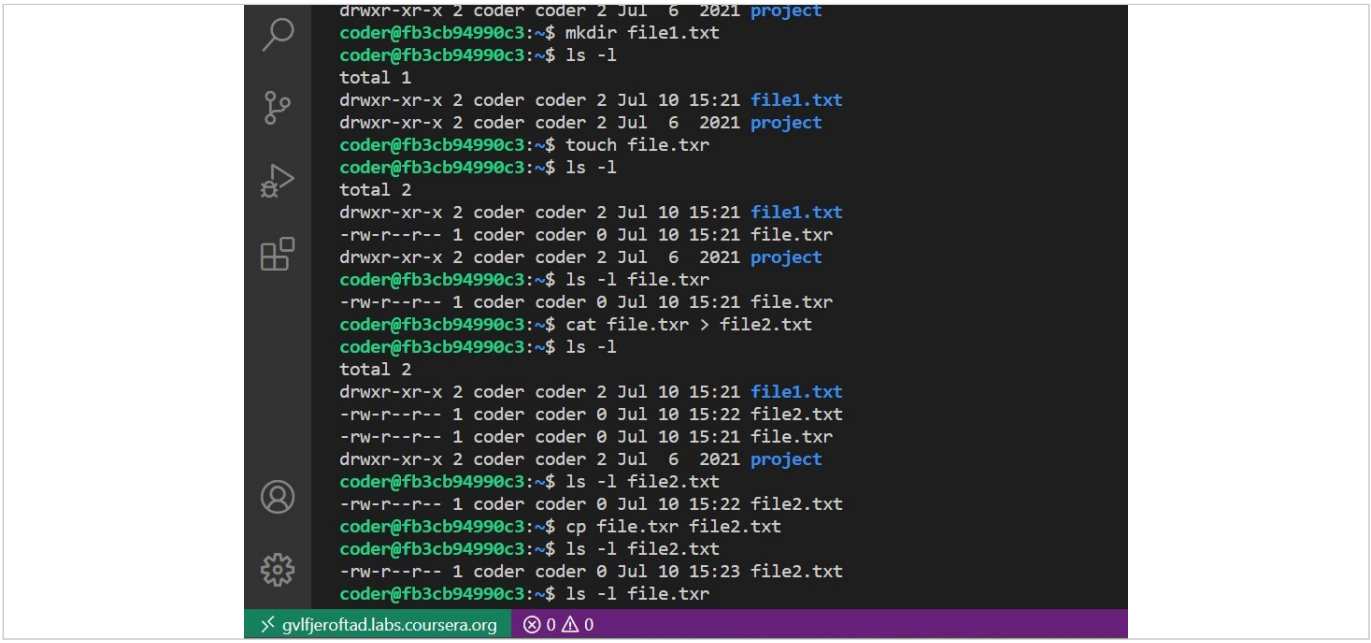

Figure 1.3 — grep -E matching multiple alternative patterns.

2. Viewing & Writing Files — cat, ls

Reading file contents, listing file details, and creating or updating files.

Command	Description
cat .profile > file.txr	Overwrites file.txr with the contents of .profile.
ls -l file_name.txt	Displays the file's details — permissions, owner, size, date — in long-listing format.
cat file.txr >> file_name.txt	Appends the contents of file.txr to file_name.txt without erasing the existing content.
cat	Used on its own, lets the user type lines of text directly into the terminal.
cat > file_name	Overwrites file_name with whatever text is typed immediately after the command.
cat <<ADDTEXT	Opens a heredoc block, allowing multiple lines of text to be entered under the label ADDTEXT.

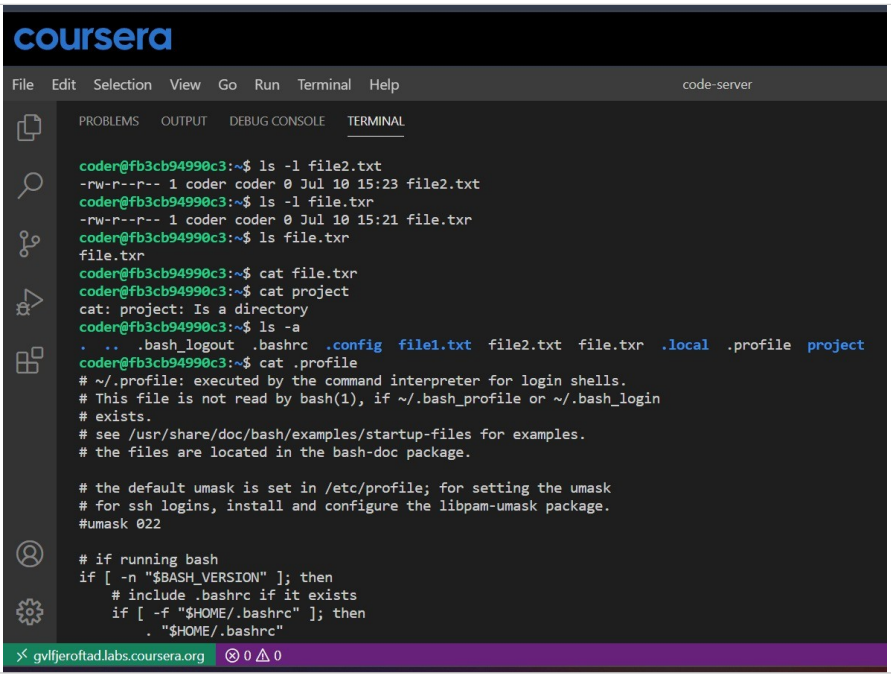

Figure 2.1 — Overwriting a file with cat and checking it with ls -l.

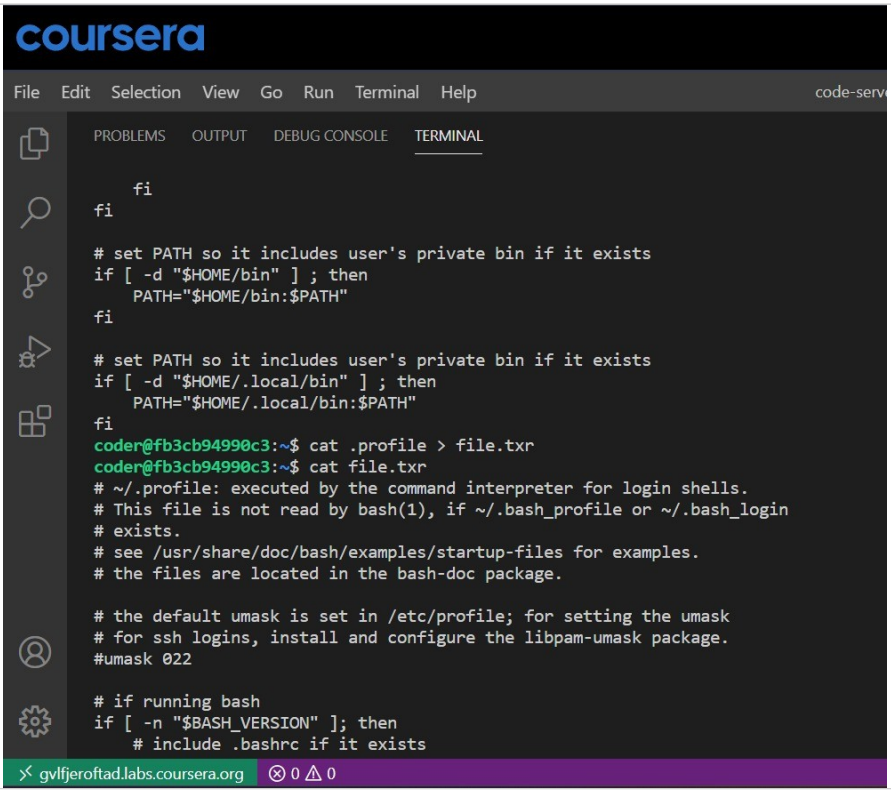

Figure 2.2 — Appending content to a file with cat >>.

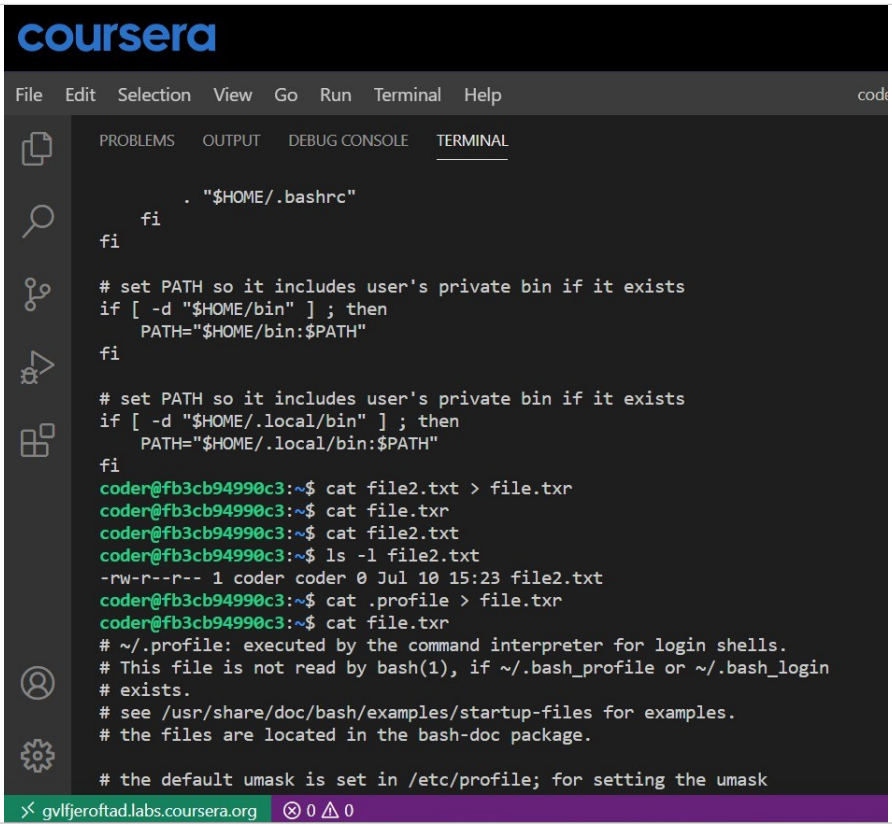

Figure 2.3 — Typing text directly into the terminal using cat.

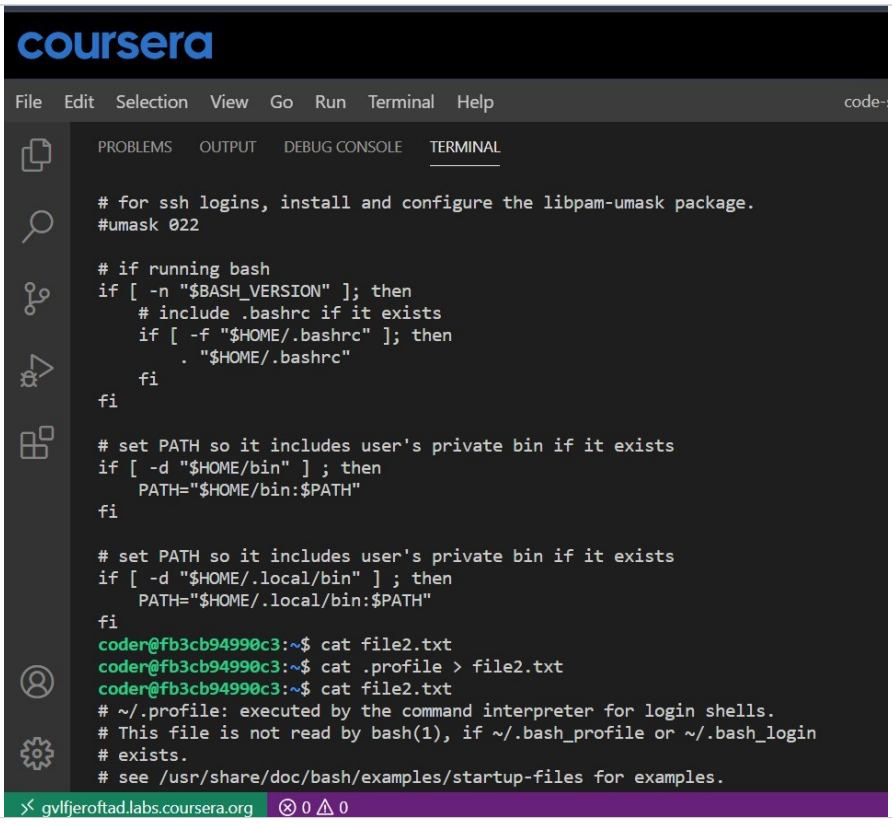

Figure 2.4 — Overwriting a file's content with cat >.

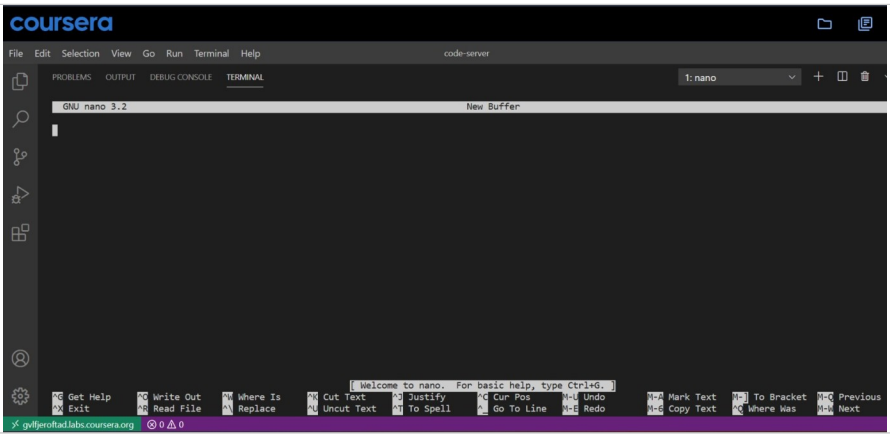

Figure 2.5 — Adding multi-line text with a cat heredoc (<<ADDTEXT).

3. File Statistics — wc

Counting lines, words, and characters, and combining results with pipes.

Command	Description
wc /etc/passwd	Displays the number of lines, words, and characters in /etc/passwd.
wc /etc/passwd | grep "passwd"	Pipes (|) the wc output into grep to filter for lines containing “passwd”.
wc /etc/passwd | tee -a file_name	Pipes the wc output into tee, which prints it on the terminal and simultaneously appends (-a) it to file_name.

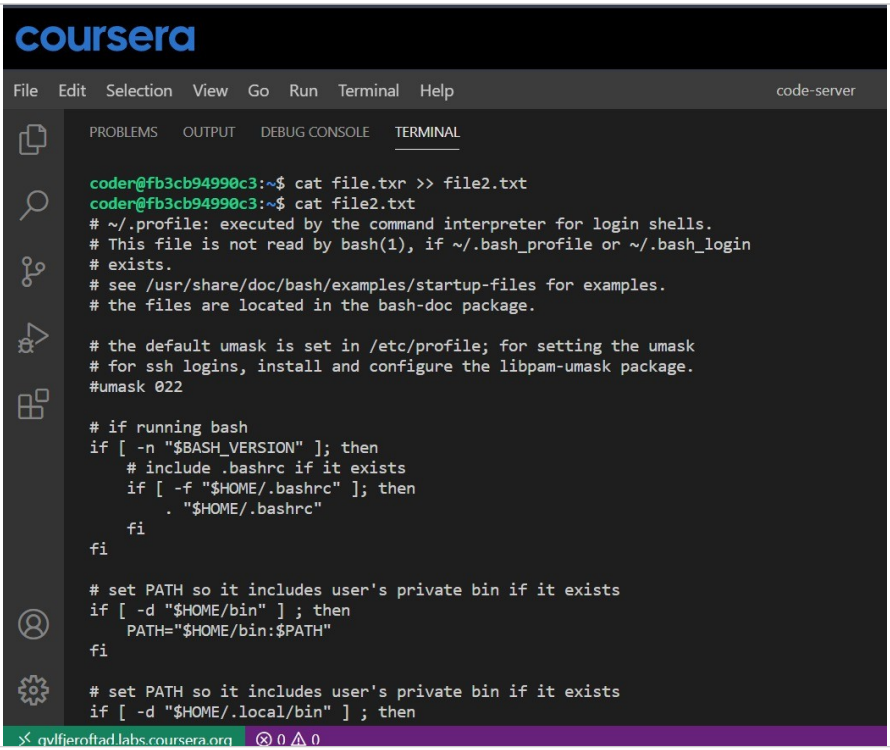

Figure 3.1 — wc showing line, word, and character counts.

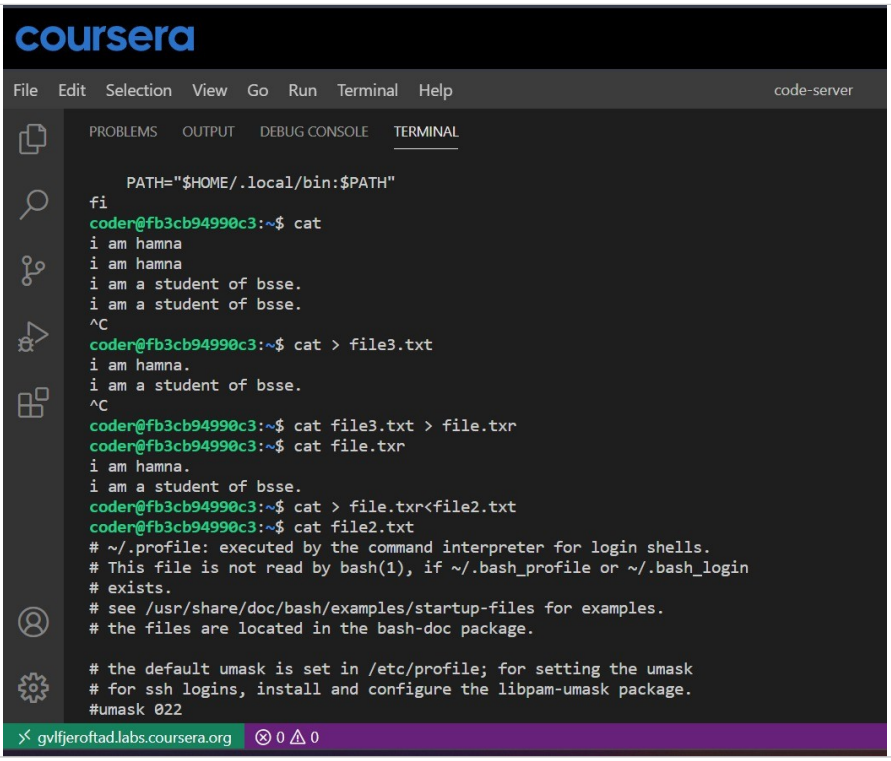

Figure 3.2 — Piping wc output into grep.

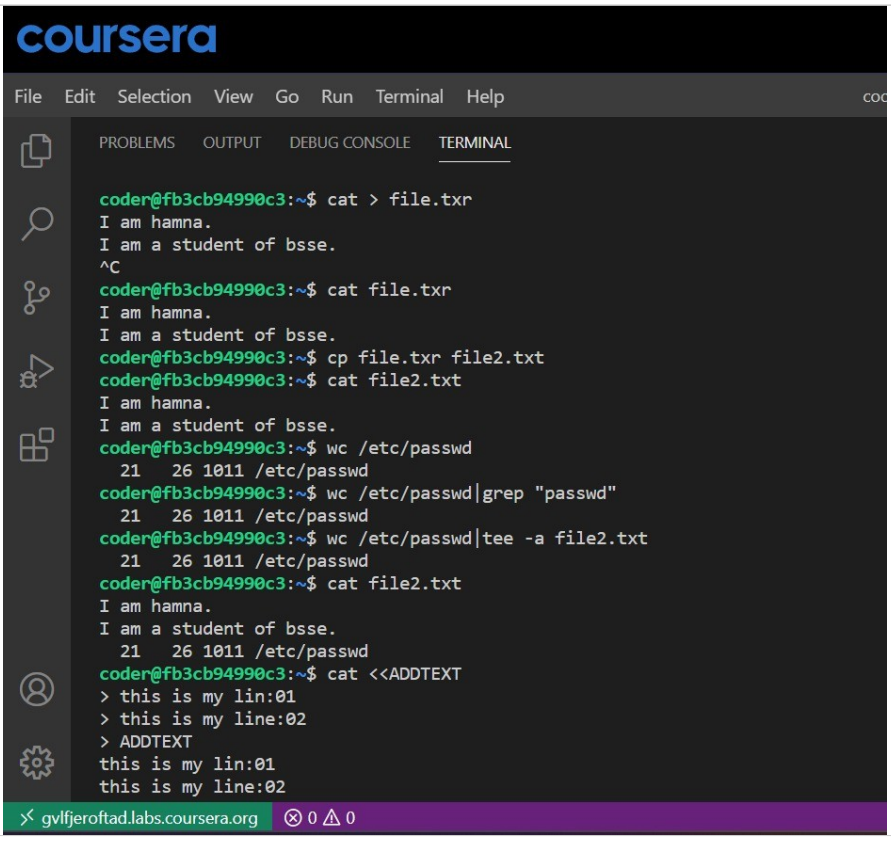

Figure 3.3 — Piping wc output into tee -a to display and append.

4. Text Editors — nano, vi

Editing files interactively from within the terminal.

Command	Description
nano	Opens the GNU nano text editor (version 3.2 shown) for direct in-terminal editing.
-- INSERT --	The nano editing mode used to type or add text; press Esc to leave this mode.
:wq	Saves changes and exits back to the terminal (used in vi/vim).
vi	Opens the vi (or vim) text editor.
vi infile.txt	Opens infile.txt inside vi for editing.

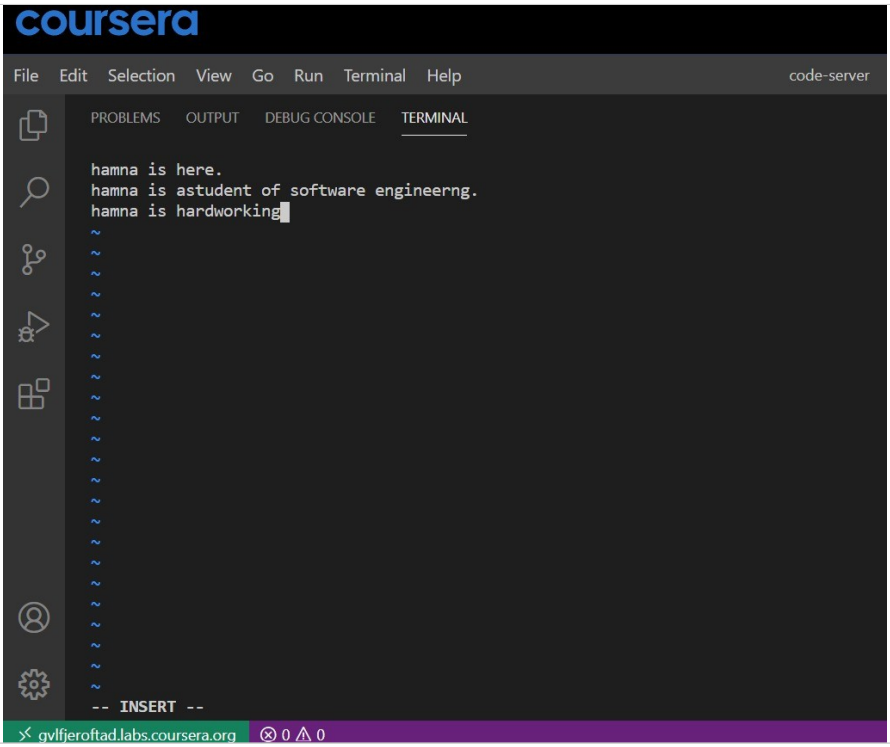

Figure 4.1 — Editing a file in nano's INSERT mode.

5. Text Processing — sed

Performing automated find-and-replace edits on file content.

Command	Description
sed -e 's/hamna/mubashra/' infile.txt	Replaces every occurrence of “hamna” with “mubashra” in infile.txt.
sed -e 's/hamna/mubashra/' infile.txt > newpoem.txt	Performs the same replacement and redirects the result into a new file, newpoem.txt.

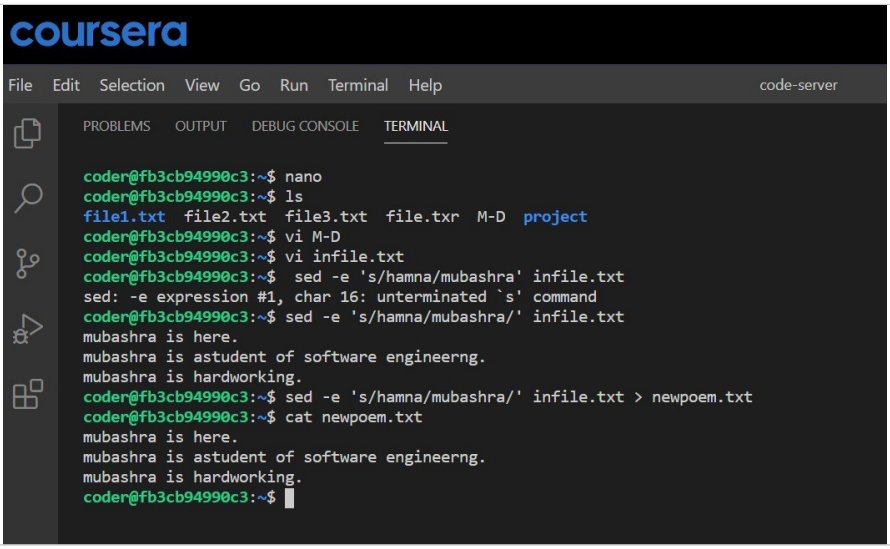

Figure 5.1 — Using vi and sed together for text substitution.

Summary

This module walked through five essential groups of Linux command-line tools, moving from locating information inside files to editing and transforming it directly in the terminal.

- Searching & Filtering (grep) — locate lines and patterns inside files using anchors, negation, and extended regular expressions.
- Viewing & Writing Files (cat, ls) — read, overwrite, append, and inspect file content and details using redirection operators (>, >>, <<).
- File Statistics (wc) — measure line, word, and character counts, and combine results with other commands through piping (|).
- Text Editors (nano, vi) — edit files interactively from within the terminal.
- Text Processing (sed) — automate find-and-replace edits and redirect results into new files.

Together, these commands form a practical toolkit for everyday Linux work: searching for information, viewing and shaping file content, gathering quick statistics, and editing or transforming text — all directly from the command line, with pipes and redirection tying them together into efficient workflows.